

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester VI Examination, 2022-23

Paper: CS-109 — Data Communication

Time: Three hours

Maximum Marks: 70

Instructions: ANSWER ANY FIVE QUESTIONS, ALL QUESTIONS CARRY EQUAL MARKS.

Question 1. Write a short note on any four of the following techniques/protocols: $[3.5 \times 4 = 14]$

- (a) Amplitude Modulation (AM)
- (b) Frequency-Division Multiplexing (FDM)
- (c) Delta Modulation (DM)
- (d) Cyclic Redundancy Check (CRC)
- (e) High-Level Data Link Control (HDLC)
- (f) Direct Sequence Spread Spectrum (DSSS)

Question 2.

- (a) What is wireless (unguided) transmission media? What are the three major types of wireless transmission media? Describe each in detail. [6]
- (b) What are circuit-switched and packet-switched networks? Discuss the differences between them. [8]

Question 3.

- (a) What is the result of scrambling the sequence 11100000000000 using the following scrambling techniques? Assume that the last non-zero signal level has been positive: [6]
 - (i) B8ZS
 - (ii) HDB3 (assume the number of non-zero pulses is odd after the last substitution)
- (b) What is the purpose of the Pulse Code Modulation (PCM) technique? Describe the components of a PCM encoder with an example. [8]

Question 4.

- (a) Describe the following mechanisms for modulating digital data into an analog signal with examples: Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK), Phase Shift Keying (PSK), and Quadrature Amplitude Modulation (QAM). [6]

- (b) Station A needs to send a message consisting of 9 packets to Station B using a sliding window (window size 3) and Go-Back-N error control strategy. All packets are ready and immediately available for transmission. If every 5th packet that A transmits gets lost (but no ACKs from B ever get lost), then what is the total number of packets that A will transmit for sending the message to B? Explain with a diagram. [8]

Question 5.

- (a) Discuss Frequency Hopping Spread Spectrum (FHSS) with a suitable diagram. [6]
- (b) A sender needs to send the four data items given in hexadecimal: 3456, ABCC, 02BC, and EEEE. Answer the following questions: [8]
- (i) Calculate the checksum at the sender site.
 - (ii) Calculate the checksum at the receiver site if the second data item is changed to ABCE.
 - (iii) Find the checksum at the receiver site if the second data item is changed to ABCE and the third data item is changed to 02BA.

Question 6.

- (a) What is data link control? Describe the key components of data link control. [6]
- (b) Why do we need flow and error control? Discuss the stop-and-wait flow control technique with an example. [8]

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